

Photoelectric effect

The emission and absorption of light was an early subject for investigation by German physicist Max Planck. As Planck attempted to formulate a theory to explain the spectral distribution of emitted light based on a classical wave model, he ran into considerable difficulty. Classical theory (Rayleigh-Jeans law) predicted that the amount of light emitted from a hot, glowing body (a black body) would increase dramatically as the wavelength decreased, whereas experiment showed that it approached zero. This discrepancy became known as the ultraviolet catastrophe. Experimental data for the radiation of light by a black body showed a distribution having a peak at a wavelength inversely proportional to temperature (Wien's law). The classical prediction for the distribution had no maximum at all. In order to reconcile theory with laboratory results, Planck was forced to develop a new model for light called the quantum model. In this model, light is emitted in small, discrete bundles or quanta.

Theoretical background*Planck's quantum theory*

By the late 1800's many physicists thought that they had explained all the main principles of the universe and discovered all the natural laws. But as scientists continued working, inconsistencies that couldn't easily be explained began showing up in some areas of study.

In 1901 Planck published his law of radiation. In it he stated that an oscillator, or any similar physical system, has a discrete set of possible energy values or levels; energies between these values never occur. Planck went on to state that the emission and absorption of radiation is associated with the transitions or jumps between two energy levels. The energy lost or gained by the oscillator is emitted or absorbed as a quantum of radiant energy, the magnitude of which is expressed by the equation:

$$E = hf \tag{1}$$

where E equals the radiant energy, f is the frequency of the radiation, and h is a fundamental constant of nature. The constant, h, became known as Planck's constant. Planck's constant was found to have significance beyond relating the frequency and energy of light, and became a cornerstone of the quantum mechanical view of the subatomic world. In 1918, Planck was awarded a Nobel prize for introducing the quantum of light.

The Photoelectric effect

In photoelectric emission, light strikes a material, causing electrons to be emitted. The classical wave model predicted that as the intensity of incident light was increased, the amplitude and thus the energy of the wave would increase. This would then cause more energetic photoelectrons to be emitted. The new quantum model, however, predicted that higher frequency light would produce higher energy photoelectrons, independent of intensity, while increased intensity would only increase the number of electrons emitted (or photoelectric current). In the early 1900s several investigators found that the kinetic energy of the photoelectrons was dependent on the wavelength, or frequency, and independent of intensity, while the magnitude of the photoelectric current, or number of electrons was dependent on the intensity as predicted by the quantum model. Einstein applied Planck's theory and explained the photoelectric effect in terms of the quantum model using his famous equation for which he received the Nobel prize in 1921:

$$E = hf = KE_{max} + W_0 \tag{2}$$

where KE_{max} is the maximum kinetic energy of the emitted photoelectrons, and W_0 is the energy needed to remove them from the surface of the material (the work function). E is the energy supplied by the quantum of light known as a photon.

The h/e experiment

A light photon with energy hf is incident upon an electron in a cathode of a vacuum tube. The electron uses a minimum W_0 of its energy to escape the cathode, leaving it with a maximum energy of KE_{max} in the form of kinetic energy. Normally the emitted electrons reach the anode of the tube, and can be measured as a photoelectric current. However, by applying a reverse potential V between the anode and the cathode, the photoelectric current can be stopped. KE_{max} can be determined by measuring the minimum reverse potential needed to stop the photoelectrons and reduce the photoelectric current to zero (but see **Note 1**). Relating kinetic energy to stopping potential gives the equation:

$$KE_{max} = eV \quad (3)$$

Therefore, using Einstein's equation,

$$hf = eV + W_0 \quad (4)$$

When solved for V , the equation becomes:

$$V = (h/e)f - (W_0/e) \quad (5)$$

If we plot V as a function of f for different frequencies of light, the V intercept is equal to $-W_0/e$ and the slope is h/e . Coupling our experimental determination of the ratio h/e with the accepted value for e , 1.602×10^{-19} C, we can determine Planck's constant, h .

Experimental setup

Set up the equipment as shown in Fig 1. Focus the light from the Mercury Vapor Light Source onto the slot in the white reflective mask on the h/e Apparatus. Tilt the Light Shield of the Apparatus out of the way to reveal the white photodiode mask inside the Apparatus (see Fig 2). Slide the Lens/Grating assembly forward and back on its support rods until you achieve the sharpest image of the aperture centered on the hole in the photodiode mask. Secure the Lens/Grating by tightening the thumbscrew.

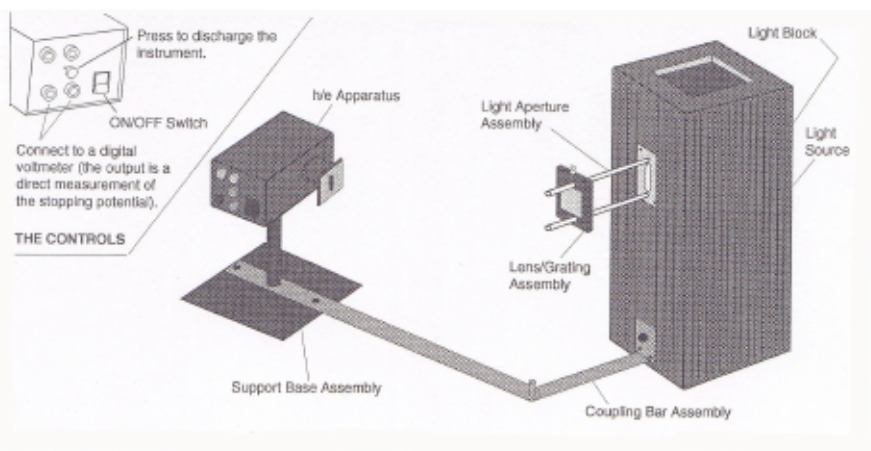


Figure 1: Photoelectric effect experimental setup

Align the system by rotating the h/e Apparatus on its support base so that the same color light that falls on the opening of the light screen falls on the window in the photodiode mask with no overlap of color from the other spectral bands. Return the Light Shield to its closed position.

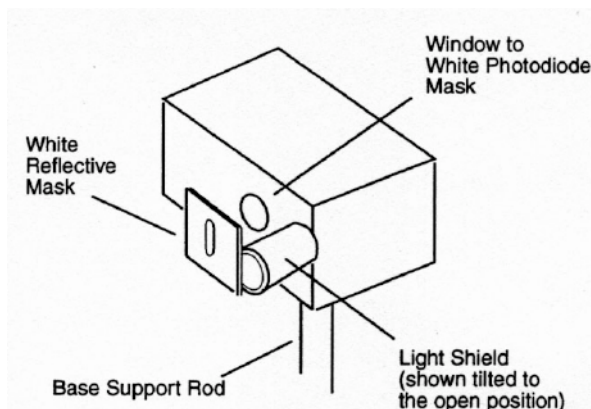


Figure 2: Detail of photodiode housing

Check the polarity of the leads from your digital voltmeter (DVM), and connect them to the OUTPUT terminals of the same polarity on the h /e Apparatus.

The h/e Apparatus includes two filters: one Green, and one Yellow. The filter frames have magnetic strips and mount to the outside of the White Reflective mask of the h /e Apparatus. Use the green and yellow filters when you're using the green and yellow spectral lines. These filters limit higher frequencies of light from entering the h /e Apparatus. This prevents ambient room light from interfering with the lower energy yellow and green light and masking the true results. It also blocks the higher frequency ultraviolet light from the higher order spectra which may overlap with lower orders of yellow and green.

Instead of applying a reverse voltage manually to find the stopping potential (when the current stops), we let the freed electrons accumulate across the face of a capacitor. This generates a reverse potential. When enough charges have accumulated, the voltage across the capacitor prevents charges from moving across the gap and the voltage stabilizes. A button on the side of the detector can discharge the capacitor, allowing charges to flow again.

Procedure

1. Turn the mercury lamp on. It might take a few seconds to light up.
2. You can see five colors in two orders of the mercury light spectrum (see Fig 3). Adjust the h /e Apparatus carefully so that only one color from the first order (the brightest order) falls on the opening of the mask of the photodiode. You will need to open the light shield (as in Fig 2) in order to see the photodiode. Close the light shield when the apparatus is adjusted for the particular wavelength that you are using.

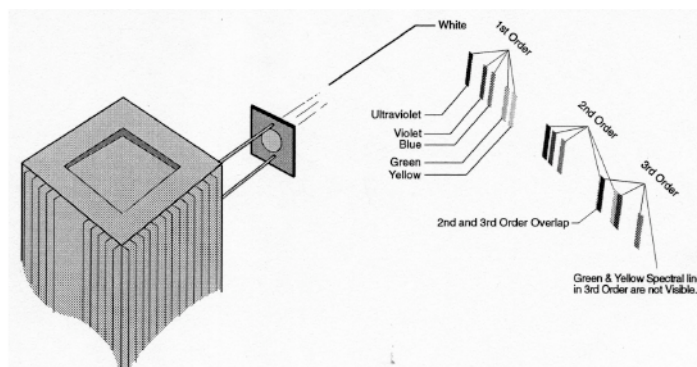


Figure 3: The Mercury lamp spectrum

3. For each color in the first order, measure the stopping potential with the multimeter and record that measurement in a table. Before reading the multimeter, press the **"PUSH TO ZERO"** button on the side panel of the h/e Apparatus to discharge any accumulated potential in the unit's electronics. This will assure the Apparatus records only the potential of the light you are measuring. (Note that the output voltage will drift with the absence of light on the photodiode.) Use the yellow and green colored filters on the Reflective Mask of the h/e Apparatus when you measure the yellow and green spectral lines.
4. Find uncertainties in your stopping potentials by repeating the measurement of each several times. Consider also the intrinsic uncertainty due to the multimeter itself.
5. Move to the second order and repeat the process. Record your results.

Analysis

1. Determine the wavelength and frequency of each spectral line. Plot a graph of the stopping potential vs. frequency. Make separate plots for your first order measurements and your second order measurements.
2. Determine the slope and y-intercept. Interpret the results in terms of the h/e ratio and the W_0/e ratio. Calculate h and W_0 .
3. In your discussion, report your values and discuss your results with an interpretation based on a quantum model of light.

The mercury spectrum

Color	Wavelength (nm)
Yellow	578.0
Green	546.1
Blue	435.8
Violet	404.7
Ultraviolet	365.5